

Sub phase -2 proposed solution

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Team ID	SWTID-2026-7624
Project name	Script-controlled ACL
Maximum marks	4 marks

Proposed Solution

The proposed solution implements a Script-Controlled Access Control List (ACL) in ServiceNow.

When a user requests access to a record, the system identifies the user's role and evaluates the defined access-control conditions. The system then decides whether the requested operation should be allowed or denied.

For example, users with the bb1 role are intended to access EEE branch records, while users with bb2, bb3, and bb4 receive Create, Write, and Delete permissions respectively. Administrators with the admin role have unrestricted access.

The Phase-1 design defines the custom roles and their CRUD responsibilities.

2.1 Key Features

1. Field-Based Restriction

The system can restrict access based on the value of a field such as Branch.

2. Role-Based Permissions

Different roles are assigned different operations according to their responsibilities.

3. Administrative Override

The admin role is provided with unrestricted access to records.

4. Dynamic Access Decision

The ACL script evaluates the conditions dynamically before granting access.

5. CRUD Control

Separate permissions are provided for:

- Create
- Read
- Write
- Delete

6. Security

Unauthorized users are prevented from accessing restricted records.

2.2 User Roles and Permissions

Role	Permission	Description
admin	Full Access	Complete access to all records
bb1	Read	View permitted EEE branch records
bb2	Create	Create new records
bb3	Write	Edit existing records
bb4	Delete	Delete existing records

2.3 Script-Controlled ACL Logic

The access decision follows the defined role and field-based security conditions.

Access Logic

User Request



Identify User Role



Check ACL Condition



Check Required Field Condition



Access Granted / Access Denied

The Phase-1 ACL design includes an administrative override and role-based access decision, followed by denial for users who do not satisfy the defined conditions

2. Proposed Solution

The proposed solution implements a Script-Controlled ACL in ServiceNow to restrict record access based on user roles and field values. The system checks the user's role and evaluates the field condition (Branch) before allowing or denying access to the records.

**Key Features**

- Field-based restriction (Branch field).
- Role-based permissions (bb1, bb2, bb3, bb4).
- Full access for admin users.
- Dynamic access using JavaScript script.
- Separate ACLs for Create, Read, Write, Delete operations.
- Easy maintenance and future scalability.

User Roles and Permissions

Role	Permission
admin	Full access to all records (all CRUD operations).
bb1	Read access (only EEE branch records).
bb2	Create access (add new records).
bb3	Write / Edit access (modify existing records).
bb4	Delete access (remove records).

Script Logic (Read ACL Example)

Condition	Action
If user has role 'admin'	Allow (full access)
If user has role 'bb1'	Allow (only EEE branch records)
Otherwise	Deny access